Assignment 4: $315 = 84 \cdot 3 + 63$
$84 = 63 \cdot 1 + 21$
$63 = 21 \cdot 3 + 0 \therefore \gcd = 21$

6.) $21 = 84 - 63(1)$
$= 84 - 315 + (84 \cdot 3)$
$= 4(84) - 1(315)$

$\therefore m = -1$
$n = 4$

2a.) if $k = \gcd(a, b)$ then $k \mid a \wedge k \mid b \therefore$
$k \mid ab$ ... wtk

2b.) is $\gcd(a,b) = \gcd(a^2, b) \rightarrow$ no its not true
proof by counter example if $a = 6 \wedge b = 4$
then $\gcd(6,4) = 2 \wedge \gcd((6^2), 4) = 4$
$\underset{36}{=}$
$\therefore$ ~~solution~~ equality does not hold

3.) $1 \times$ last dig, $x \cdot 5 + y$: assume $5b + a \equiv 0 \mod 7$

when $n = 7k$, $n = 10a + b \therefore 7k = 10a + b$

$35k = 50a + 5b$
$35k - 49a = 5b + a$
$7(5k - 7a) = 5b + a$
$\therefore 5b + a \equiv 0 \mod 7$

if $5b + a$ is a multiple of 7
$5 \times n = 5(10a + b) = 50a + 5b = 49a + (5b$

"as $49a$ is $\equiv 0 \mod 7$ and $5b + a \equiv 0 \mod 7$, then the statement is true."